

Commutators and functions of operators:

- 1.) When can we make the replacement $e^{-i\hat{H}t/\hbar} \rightarrow e^{-iEt/\hbar}$?

Solution: When the energy is well defined, meaning the operator above acts on an eigenstate:

$$e^{-i\hat{H}t/\hbar}|E_n\rangle = e^{-iE_nt/\hbar}|E_n\rangle$$

- 2.) Describe to your neighbor how the exponential of an operator even makes sense. For conversational purposes, let us consider a specific example: why can I write $e^{-i\hat{H}t/\hbar}|E_n\rangle = e^{-iE_nt/\hbar}|E_n\rangle$? Here $|E_n\rangle$ refers to the n^{th} energy eigenstate.

Solution: Consider expanding the exponential:

$$\begin{aligned} e^{-i\hat{H}t/\hbar}|E_n\rangle &= \left(1 - i\frac{\hat{H}t}{\hbar} - \frac{\hat{H}^2t^2}{2\hbar^2} + i\frac{\hat{H}^3t^3}{6\hbar^3} + \dots\right)|E_n\rangle \\ &= \left(1 - i\frac{E_nt}{\hbar} - \frac{E_n^2t^2}{2\hbar^2} + i\frac{E_n^3t^3}{6\hbar^3} + \dots\right)|E_n\rangle \\ &= e^{-iE_nt/\hbar}|E_n\rangle \end{aligned}$$

Acting with several orders of the Hamiltonian on an energy eigenstate makes perfect sense.

- 3.) Your friend is working with two operators $\hat{A}$ and $\hat{B}$ on a complex vector space V , but he doesn't really know much about this vector space. He tells you $[\hat{A}, \hat{B}] = ic\hat{I}$, where $\hat{I}$ is the identity, and c is a constant. What, if anything, can you tell your friend about the dimensionality of his vector space?

Solution: Recall that the trace of a product of matrices is independent of the order in which the matrices are multiplied, assuming the trace is well-defined (including finite): $Tr(\hat{A}\hat{B}) = Tr(\hat{B}\hat{A})$. So that means

$$\begin{aligned} Tr(\hat{A}\hat{B} - \hat{B}\hat{A}) &= Tr(c\hat{I}) \\ Tr(\hat{A}\hat{B}) - Tr(\hat{B}\hat{A}) &= cTr(\hat{I}) \\ 0 &= cTr(\hat{I}) \end{aligned}$$

And we have arrived at a contradiction. This means something in our assumption was wrong. If a matrix is infinite dimensional, then its trace is ill-defined (it is defined, but it can diverge such that the difference of two diverging terms can certainly be non-zero). So, the vector space is infinite dimensional. This is the case for observables like $\hat{x}$ and $\hat{p}_x$.

- 4.) Is $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}$? Convince your neighbor. Hint: only consider cross terms up to order $\hat{A}\hat{B}$.

Solution: In general, no. They are only equal in special cases. Consider the expansion of the exponentials:

$$e^{\hat{A}}e^{\hat{B}} = \left(1 + \hat{A} + \frac{\hat{A}^2}{2} + \frac{\hat{A}^3}{6} + \dots\right)\left(1 + \hat{B} + \frac{\hat{B}^2}{2} + \frac{\hat{B}^3}{6} + \dots\right)$$

$$= 1 + \hat{A} + \hat{B} + \hat{A}\hat{B} + \frac{\hat{A}^2}{2} + \frac{\hat{B}^2}{2} + \frac{\hat{A}\hat{B}^2}{2} + \frac{\hat{B}\hat{A}^2}{2} + \frac{\hat{A}^3}{6} + \frac{\hat{B}^3}{6} + \dots$$

Whereas

$$\begin{aligned} e^{\hat{A}+\hat{B}} &= \left(1 + (\hat{A} + \hat{B}) + \frac{(\hat{A} + \hat{B})^2}{2} + \frac{(\hat{A} + \hat{B})^3}{6} + \dots \right) \\ &= 1 + \hat{A} + \hat{B} + \frac{\hat{A}\hat{B}}{2} + \frac{\hat{B}\hat{A}}{2} + \frac{\hat{A}^2}{2} + \frac{\hat{B}^2}{2} + \frac{1}{6} \left(\hat{A}\hat{B}^2 + \hat{B}\hat{A}^2 + \hat{A}^2\hat{B} + \hat{B}^2\hat{A} + \hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}\hat{B} + \frac{\hat{A}^3}{6} + \frac{\hat{B}^3}{6} \right) + \dots \end{aligned}$$

From here it is clear that to zeroth and first order in the operators, the two expressions agree. But to second order they disagree by

$$e^{\hat{A}}e^{\hat{B}} - e^{\hat{A}+\hat{B}} = \frac{1}{2}(\hat{A}\hat{B} - \hat{B}\hat{A}) + O(\text{op}^3)$$

In fact, the higher orders become very complicated for general commutation relations. The total formula relating the two exponential expressions is called the Baker-Campbell-Hausdorff formula. In the special case where the commutator $[\hat{A}, \hat{B}]$ is a complex number (times the identity), the higher orders drop out and we have $e^{\hat{A}}e^{\hat{B}} - e^{\hat{A}+\hat{B}} \rightarrow \frac{1}{2}(\hat{A}\hat{B} - \hat{B}\hat{A})$

Of course, if the two operators commute, then even this one term is zero.

5.) Is $e^{\hat{A}}e^{-\hat{A}} = \hat{I}$? Convince your neighbor.

Solution: Yes. Since $\hat{A}$ commutes with itself (!!!) we can write

$$\begin{aligned} e^{\hat{A}}e^{-\hat{A}} - e^{\hat{A}+(-\hat{A})} &= \frac{1}{2}(\hat{A}\hat{A} - \hat{A}\hat{A}) = 0 \\ e^{\hat{A}}e^{-\hat{A}} &= e^{\hat{A}+(-\hat{A})} = \hat{I} \end{aligned}$$

6.) Suppose that $[\hat{A}, \hat{B}] = c\hat{I}$, where c is a constant (possibly complex). Calculate $[e^{\hat{B}}, \hat{A}]$. Here you may use that $[\hat{B}^n, \hat{A}] = n[\hat{B}, \hat{A}]\hat{B}^{n-1}$, although it might be useful to convince yourself this is true for the first few orders.

Solution: We can write $[e^{\hat{B}}, \hat{A}] = \left[\sum_m \frac{\hat{B}^m}{m!}, \hat{A} \right] = \sum_m \frac{1}{m!} [\hat{B}^m, \hat{A}]$. Now, note that

$$\begin{aligned} [\hat{B}^n, \hat{A}] &= \hat{B}^n\hat{A} - \hat{A}\hat{B}^n = \hat{B}^{n-1}\hat{B}\hat{A} - \hat{A}\hat{B}^n = \hat{B}^{n-1}(\hat{A}\hat{B} + [\hat{B}, \hat{A}]) - \hat{A}\hat{B}^n \\ &= \hat{B}^{n-2}(\hat{B}\hat{A}\hat{B} + \hat{B}[\hat{B}, \hat{A}]) - \hat{A}\hat{B}^n = \hat{B}^{n-2}((\hat{A}\hat{B} + [\hat{B}, \hat{A}])\hat{B} + \hat{B}[\hat{B}, \hat{A}]) - \hat{A}\hat{B}^n \\ &= \hat{B}^{n-2}(\hat{A}\hat{B}^2 + 2\hat{B}[\hat{B}, \hat{A}]) - \hat{A}\hat{B}^n \\ &\vdots \\ [\hat{B}^n, \hat{A}] &= n[\hat{B}, \hat{A}]\hat{B}^{n-1} \end{aligned}$$

Finally, we have

$$[e^{\hat{B}}, \hat{A}] = \sum_m \frac{1}{m!} m [\hat{B}, \hat{A}] \hat{B}^{m-1} = \sum_m \frac{1}{(m-1)!} (-c\hat{I}) \hat{B}^{m-1} = -ce^{\hat{B}}$$

Spin

7.) Consider the operator:

$$\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$$

Where $\hat{\vec{S}} = \hbar \hat{\vec{\sigma}}/2$ and $\hat{\vec{\sigma}}$ is a vector of the Pauli matrices.

(a) Give a physical interpretation of this operator.

(b) What are this operator's eigenvalues?

Solution:

(a) Solution: This is the spin operator along some direction $\hat{n}$, which is at the angles θ, ϕ .

(b) Starting from $\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$ we can write this as a matrix:

$$\begin{aligned} \hat{S}_{\hat{n}} &= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta) \cos(\phi) - i \sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) + i \sin(\theta) \sin(\phi) & -\cos(\theta) \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta) e^{-i \sin(\phi)} \\ \sin(\theta) e^{i \sin(\phi)} & -\cos(\theta) \end{pmatrix} \end{aligned}$$

So, the eigenvalues are given by

$$\left| \begin{pmatrix} \frac{\hbar}{2} \cos(\theta) - \lambda & \frac{\hbar}{2} \sin(\theta) e^{-i\phi} \\ \frac{\hbar}{2} \sin(\theta) e^{i\phi} & -\frac{\hbar}{2} \cos(\theta) - \lambda \end{pmatrix} \right| = -\left(\frac{\hbar}{2}\right)^2 \cos^2(\theta) + \lambda^2 - \left(\frac{\hbar}{2}\right)^2 \sin^2(\theta) = 0 \Rightarrow \lambda = \pm \frac{\hbar}{2}$$

This fits our expectation, since this is just a spin operator of the same size/spin as our familiar old $\hat{S}_z$, just oriented along a new axis.

8.) Consider the state $\frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3i \end{pmatrix}$.

(a) What is the physical interpretation of this state? What are the possible outcomes and corresponding probabilities for making $\hat{S}_z$ measurements?

(b) Explain how one can produce such a state in the lab. HINT: what equipment / setup would you need and how would you orient it?

You can use that the eigenstates of

$$\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$$

are (in the S_z basis)

$$\begin{aligned} |+\rangle &= \cos(\theta/2) |\uparrow\rangle + \sin(\theta/2) e^{i\phi} |\downarrow\rangle \\ |-\rangle &= \sin(\theta/2) |\uparrow\rangle - \cos(\theta/2) e^{i\phi} |\downarrow\rangle \end{aligned}$$

Solution:

(a) This is a state that has a well-defined spin projection along SOME axis, but not along x, y, or z. The probability of a measurement of $\hat{S}_z$ yielding $+\hbar/2$ is 4/13, and of yielding $-\hbar/2$ is 9/13.

(b) As part of homework, students are finding the eigenstates to the $\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$ matrix. They should find,

$$\begin{aligned}|+n\rangle &= \cos(\theta/2) |\uparrow\rangle + \sin(\theta/2) e^{i\phi} |\downarrow\rangle \\ |-n\rangle &= \sin(\theta/2) |\uparrow\rangle - \cos(\theta/2) e^{i\phi} |\downarrow\rangle\end{aligned}$$

If we select $|+n\rangle$ from a magnet oriented along the $+n$ direction, we need the coefficient of $|\downarrow\rangle$ to be purely imaginary. So, we set $\phi = \pi/2$. The ratio of the coefficients of the two orthogonal basis states is $\frac{\sin(\theta/2)}{\cos(\theta/2)} = \tan(\theta/2) = \frac{3}{2} \Rightarrow \theta = 2 \tan^{-1}(3/2) = 113^\circ$.

Spin Precession

- 9.) Using classical analogues of spin (circulating charge), explain why we would expect the Hamiltonian of a spin to involve the spin's interaction with a magnetic field. Write an expression for the Hamiltonian in terms of other constants. What is different about this compared to the classical case?

Solution: A current loop is circulating current and is a classical analogue of spin. A current loop produces a magnetic field that will tend to align with an external magnetic field. The energy of a magnetic moment in an external field is:

$$\hat{H} = -\hat{\vec{\mu}} \cdot \vec{B}$$

Here, the magnetic moment is $\hat{\vec{\mu}} = -\left(\frac{eg}{2m}\right)\hat{\vec{S}}$, where the “-“ is from the charge of the electron, and g is the Lande g -factor for spin, which is very close to 2. Also, $\frac{e}{2m} = \mu_B$ is called the Bohr magneton. Altogether, we have:

$$\hat{H} = -\hat{\vec{\mu}} \cdot \vec{B} = \left(\frac{eg}{2m}\right)\hat{\vec{S}} \cdot \vec{B}$$

10.) Suppose you prepare a state $|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$ at time $t = 0$.

- (a) Immediately after this state is prepared (after a time much smaller than characteristic time scales for the system), find the probability that a measurement of spin projection onto the z axis yields $+\hbar/2$.
(b) Now let us switch on a magnetic field $\vec{B} = B_0\hat{i}$. Write down the time-dependent state.
(c) Find the probability that a measurement of the spin projection on the z axis will yield $+\hbar/2$, as a function of time. Hint: you may use the fact that $\hat{H} = -\hat{\vec{\mu}} \cdot \vec{B} = \left(\frac{eg}{2m}\right)\hat{\vec{S}} \cdot \vec{B}$, where $g_e \approx 2$, and note (cryptically) the units of $\left(\frac{eB_0}{m}\right)$.

Solution:

$$(a) P_{+z} = \left| \langle + | \psi \rangle \right|^2 = \left| (1, 0) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \right|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} = 50\%$$

(b) We will first do this in the basis that diagonalizes the Hamiltonian:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi\rangle = e^{-i\omega\hat{S}_x t/\hbar}|\psi\rangle = e^{-i\omega\hat{S}_x t/\hbar} \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} (|+\rangle_x + |-\rangle_x) + \frac{i}{\sqrt{2}} (|+\rangle_x - |-\rangle_x) \right)$$

$$|\psi(t)\rangle = \frac{1}{2} \left((e^{-i\omega t/2} |+\rangle_x + e^{i\omega t/2} |-\rangle_x) + i(e^{-i\omega t/2} |+\rangle_x - e^{i\omega t/2} |-\rangle_x) \right)$$

This is a fine form for the time-dependent state. But since we know we eventually want to take the inner product with eigenstates of $\hat{S}_z$, we will go ahead and express this in terms of eigenstates of $\hat{S}_z$:

$$|\psi(t)\rangle = \frac{1}{2\sqrt{2}} \left((e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) + e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) + i(e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) - e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) \right)$$

(c)

$$\begin{aligned} P_{+z} &= \left| \langle + | \psi \rangle \right|^2 \\ &= \left| \langle + | \frac{1}{2\sqrt{2}} \left((e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) + e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) + i(e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) - e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) \right) \right|^2 \\ &= \frac{1}{8} \left| \left((e^{-i\omega t/2} + e^{i\omega t/2}) + i(e^{-i\omega t/2} - e^{i\omega t/2}) \right) \right|^2 \\ &= \frac{1}{8} \left((e^{-i\omega t/2} + e^{i\omega t/2}) + i(e^{-i\omega t/2} - e^{i\omega t/2}) \right) \left((e^{i\omega t/2} + e^{-i\omega t/2}) - i(e^{i\omega t/2} - e^{-i\omega t/2}) \right) \\ &= \frac{1}{8} \left((2 + e^{-i\omega t} + e^{i\omega t}) + (2 - e^{-i\omega t} - e^{i\omega t}) - i(e^{i\omega t} - e^{-i\omega t}) + i(e^{-i\omega t} - e^{i\omega t}) \right) \\ &= \frac{1}{8} \left(4 + \frac{2}{i} (e^{i\omega t} - e^{-i\omega t}) \right) = \frac{1}{2} \left(1 + \frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \right) \\ &= \frac{1}{2} (1 + \sin(\omega t)) \end{aligned}$$

Note that at $t = 0$, $P \rightarrow \frac{1}{2}$, which makes sense, since the original state was an eigenstate of $\hat{S}_y$.

11.) Suppose a particle is prepared in an eigenstate of $\hat{S}_z$ at time zero: $|\psi(t=0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. But right at time zero, a magnetic field is switched on with $\vec{B} = B_0 \hat{z}$.

- Use intuition to argue whether any of the quantities $\langle \hat{S}_x(t) \rangle$, $\langle \hat{S}_y(t) \rangle$, or $\langle \hat{S}_z(t) \rangle$ change in time.
- Write out the time-dependent state.
- Check your answer to (a) with explicit calculation.

Solution:

(a) One can consider the precessing vector model to see that there can be no non-trivial time evolution of the expectation value of $\langle \hat{S}_x(t) \rangle$: the spin vector (or rather two constituent vectors making it up) precess around the x axis. Their projections along the x axis cannot change in time. The same argument cannot be made for other axes. It seems likely that $\langle \hat{S}_z(t) \rangle$ will change in time, since it starts out at a maximum and the vectors precess around a different axis. However, since the initial expectation value of $\langle \hat{S}_y(t) \rangle$ is zero, it is not obvious whether this will change in time or whether the y components of the vectors will always cancel.

(b) The Hamiltonian for a magnetic moment in a magnetic field is $\hat{H} = -\vec{\mu} \cdot \vec{B} = \left(\frac{eg}{2m}\right) \hat{S} \cdot \vec{B} \rightarrow \left(\frac{e}{m}\right) \hat{S} \cdot \vec{B}$.
 $\vec{B} \rightarrow \left(\frac{eB_0}{m}\right) \hat{S}_x \equiv \omega \hat{S}_x$.

To write the time-dependent state, we act on the initial state with the time-evolution operator:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(t=0)\rangle = e^{-i\omega t \hat{S}_x/\hbar} |\psi(t=0)\rangle$$

Because the initial state is not an eigenstate of $\hat{S}_x(t)$, we cannot insert eigenvalues for the operator, unless we write the initial state as a superposition of $\hat{S}_x(t)$ eigenstates. That is $|\psi(t=0)\rangle = |\uparrow\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$

$$\frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} (|+\rangle_x + |-\rangle_x).$$

Thus, we can write:

$$|\psi(t)\rangle = e^{-i\omega t \hat{S}_x/\hbar} |\psi(t=0)\rangle = \frac{1}{\sqrt{2}} (e^{-i\omega t/2} |+\rangle_x + e^{i\omega t/2} |-\rangle_x) = \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix}.$$

(c) Given the spinor from (b):

$$\begin{aligned} \langle \hat{S}_x(t) \rangle &= \langle \psi(t) | \hat{S}_x | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} -i \sin(\omega t/2) \\ \cos(\omega t/2) \end{pmatrix} = 0 \end{aligned}$$

As we reasoned in part (a).

$$\begin{aligned} \langle \hat{S}_y(t) \rangle &= \langle \psi(t) | \hat{S}_y | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} -\sin(\omega t/2) \\ i \cos(\omega t/2) \end{pmatrix} = 0 \end{aligned}$$

We did not know for sure that this would be the case, but we suspected it could be. Finally,

$$\begin{aligned} \langle \hat{S}_z(t) \rangle &= \langle \psi(t) | \hat{S}_z | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} \cos(\omega t/2) \\ i \sin(\omega t/2) \end{pmatrix} = (\hbar/2)(\cos^2(\omega t/2) - \sin^2(\omega t/2)) = (\hbar/2) \cos(\omega t) \end{aligned}$$

We knew there had to be non-trivial time dependence for this. Note also that the boundary condition matches, in that the state is spin-up at time zero.

Quantum Zeno effect (setup for Problem 2, HW3)

12.) Consider a single qubit evolving under the following Hamiltonian:

$$\hat{H} = -\mu_B B \hat{\sigma}_z$$

Where $\mu_B = \hbar e/2m_e$ is the Bohr magneton.

Starting from an initial state, at time $t = 0$:

$$|\psi_0\rangle = |+\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}}$$

The qubit evolves for a time T followed by a measurement of $\hat{\sigma}_x$.

- Give an example of a physical system with such a Hamiltonian
- What convention are we using for the notation of spin? That is, what does $|+\rangle$ represent? What do $|\uparrow\rangle$ and $|\downarrow\rangle$ represent?
- What is the probability at time T to produce a measurement outcome of $\sigma_x = +1$?

Solution:

- A spin-1/2 particle (an electron) inside a uniform, constant (time-independent) magnetic field in the z -direction
- $|+\rangle$ is the eigenvector for $\hat{\sigma}_x$ with eigenvalue $+1$. $|\uparrow\rangle$ and $|\downarrow\rangle$ are the eigenvectors of $\hat{\sigma}_z$ with eigenvalues $+1$ and -1 , respectively.
- First, time-evolve the initial state $|\psi_0\rangle$ in the $\hat{\sigma}_z$ basis. Then, represent the time-evolved state $|\psi(t)\rangle$ in the $\hat{\sigma}_x$ basis to calculate the probability of measuring $\sigma_x = +1$.

$$\begin{aligned} |\psi(t)\rangle &= e^{-i\hat{H}t/\hbar} |\psi_0\rangle \\ |\psi(t)\rangle &\doteq e^{\frac{i\mu_B B t \hat{\sigma}_z}{\hbar}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\ |\psi(t)\rangle &\doteq \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\omega t} \\ e^{-i\omega t} \end{pmatrix} \end{aligned}$$

Where we used $\omega = \mu_B B / \hbar$ in the last step. This is the time-evolved state in the $\hat{\sigma}_z$ basis. We can write it in the $\hat{\sigma}_x$ basis by noting that

$$|+\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|\uparrow\rangle - |\downarrow\rangle}{\sqrt{2}} \text{ such that } |\uparrow\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |\downarrow\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}, \text{ and remembering that } e^{i\omega t} = \cos \omega t + i \sin \omega t \text{ and } e^{-i\omega t} = \cos \omega t - i \sin \omega t.$$

$$\begin{aligned} |\psi(t)\rangle &\doteq \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\omega t} \\ e^{-i\omega t} \end{pmatrix} \doteq \frac{1}{\sqrt{2}} e^{i\omega t} |\uparrow\rangle + \frac{1}{\sqrt{2}} e^{-i\omega t} |\downarrow\rangle \\ |\psi(t)\rangle &= \frac{1}{\sqrt{2}} (\cos \omega t + i \sin \omega t) |\uparrow\rangle + \frac{1}{\sqrt{2}} (\cos \omega t - i \sin \omega t) |\downarrow\rangle \\ |\psi(t)\rangle &= \frac{1}{\sqrt{2}} (\cos \omega t + i \sin \omega t) \left(\frac{|+\rangle + |-\rangle}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} (\cos \omega t - i \sin \omega t) \left(\frac{|+\rangle - |-\rangle}{\sqrt{2}} \right) \end{aligned}$$

Re-arranging terms:

$$|\psi(t)\rangle = \cos \omega t |+\rangle + i \sin \omega t |-\rangle$$

From this we can see that the outcomes and probabilities of taking a measurement in $\hat{\sigma}_x$ are $P(+1) = \cos^2 \omega t$ and $P(-1) = \sin^2 \omega t$.